



DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Experiment No.: 9

Interfacing Ultrasonic Sensor HC-SR04 with STM32F4xx.

Name: _____

Roll No.: _____ *Batch:* _____

Date of Performance: _____

Date of Assessment: _____

Particulars	Marks
Marks for Regularity (05)	
Marks for Performance (15)	
Understanding (Viva) (05)	
Total (25)	
Signature of Course Teacher	

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Experiment: 9

AIM: Interfacing Ultrasonic Sensor HC-SR04 with STM32F4xx.

OBJECTIVE:

1. To interface the HC-SR04 ultrasonic distance sensor with the STM32F407 microcontroller and measure the distance of an object by generating a precise 10 μ s trigger pulse and measuring the width of the returned echo pulse using a hardware timer in input capture mode.
2. To understand the working principle of the HC-SR04.

APPARATUS:

1. STM32F407 module development board.
2. Ultrasonic sensor – HC-SR04
3. Connecting wire.
4. Adapter.

THEORY:

The Ultrasonic Sensor is an electronic device that calculates distance by emitting sound waves and collecting their echoes.

It can measure objects from up to 4.5 meters away, which makes it a versatile instrument for correctly measuring both short and long distances without making contact with the target object, which is critical in many applications such as obstacle avoidance systems in robotics or autonomous cars.

The sensor features adjustable pulse widths, allowing for higher resolution measurements when set at lower levels though this comes at the cost of range accuracy diminishing as a result, however, due to its affordability compared to other forms of distance sensing technologies makes it incredibly suitable where budget constraints are paramount factor within projects evaluating solutions using ultrasonic technology.

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Ultrasonic sensors are used in parking technologies and anti-collision safety systems. Ultrasonic proximity sensors are also employed in automated obstacle detection systems and factory engineering.

Compared to infrared (IR) sensors in proximity sensing applications, ultrasonic sensors are less susceptible to interference from smoke, gases, and other airborne particles (although the physical component is subject to variables such as heat).

Ultrasonic sensors are not just used to measure liquid levels in containers, they also enable doctors to view inside organs, identify cancer, and monitor babies in the womb.

Working Principle of Ultrasonic Sensor?

An ultrasonic sensor is a type of electronic sensor that uses ultrasonic waves to determine the distance between two objects and converts the reflected sound into electrical signals.

The working principle of an ultrasonic sensor is to measure distance using ultrasound, which travels faster than sound that is audible. This sensor consists of two major components a transmitter that generates sound waves via a piezoelectric crystal and a receiver that detects the reflected ultrasonic waves.

While some sensors have separate sound emitters and receivers, the two functions can be combined into a single device by using an ultrasonic element that alternates between sending and receiving signals in a continuous loop.

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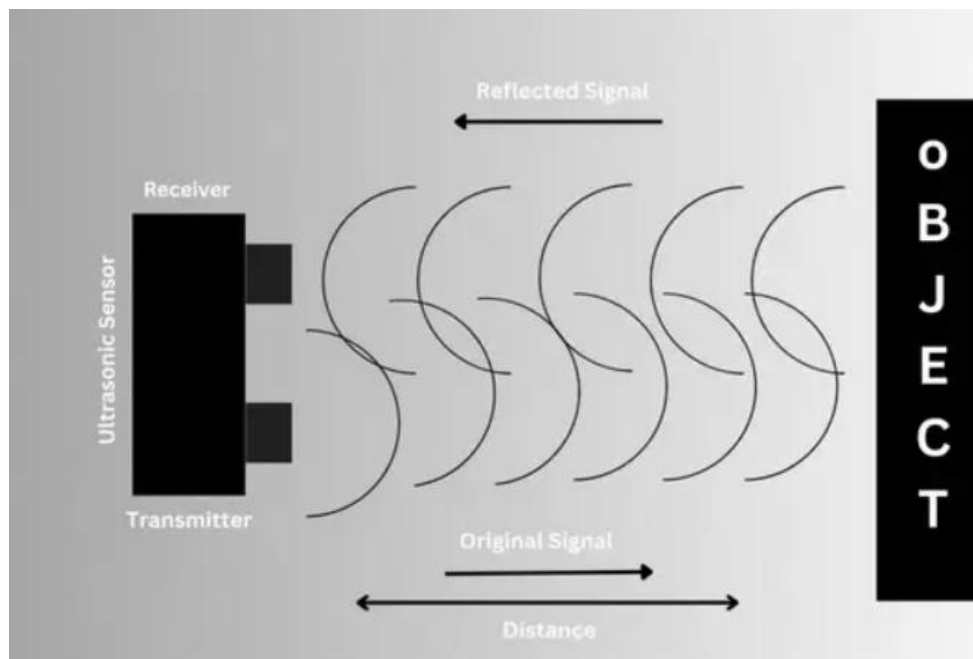
The transmitter of the module transmits an ultrasonic sound. This sound will be reflected if an object is present in front of the ultrasonic sensor. The reflected sound is received by the receiver present in the same module.

An ultrasonic signal is propagated by a wave at an angle of 30°. The above-depicted Figure illustrates how the ultrasonic signal propagates from the transmitter. Measuring angles should be at least 15° for maximum accuracy.

In this case, external objects that fall under this measurement angle interfere with determining the distance to the desired object.

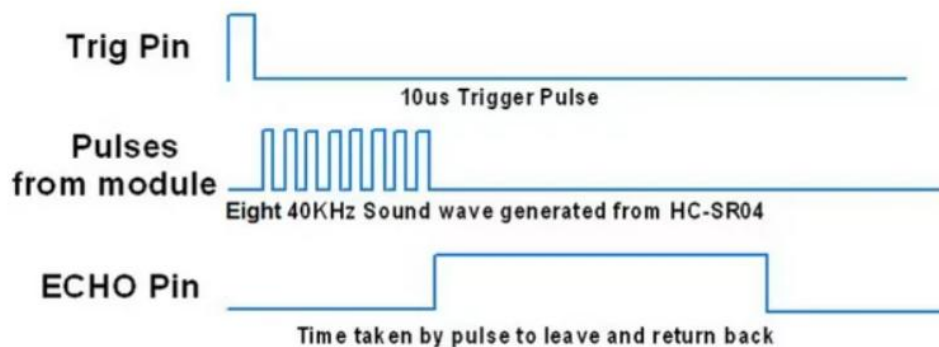
The distance is calculated by measuring the ultrasonic sound's travel time and speed.

$$\text{Distance} = \text{Time} \times \text{Speed of sound} / 2$$



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Ultrasonic HC-SR04 module Timing Diagram



Specifications:

- Supply voltage +5 V;
- Consumption in silent mode 2 mA;
- Consumption at work of 15 mA;
- Measurement range - 2 to 400 cm;
- Effective measuring angle 15°;
- The dimensions are 45×20×15 mm.

Applications of Ultrasonic Sensor HC-SR04:

- Ultrasonic Anemometers.
- Tide Gauge.
- Tank level.
- UAV navigation.
- Web-guiding system.

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1. Using the STM32CubeIDE.

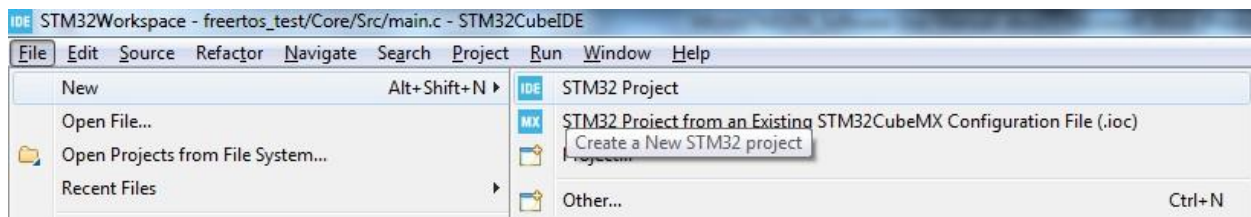
1.1. Creating/Choosing a Workspace.

When you open the STM32CubeIDE for the first time, you will be directed to create a workspace for your projects. You can choose the default location for the project provided by the IDE (which will generally be located in the **users** folder in C: drive) or you can assign a different folder for the workspace. All your projects will be stored and referenced from this workspace folder. STM32CubeIDE is a eclipse based IDE and so you can have multiple projects in your projects window. Only one project can be active at one time. The Active project will get compiled or debugged or downloaded when you use the shortcut keys for the Compile, Debug or RUN actions.

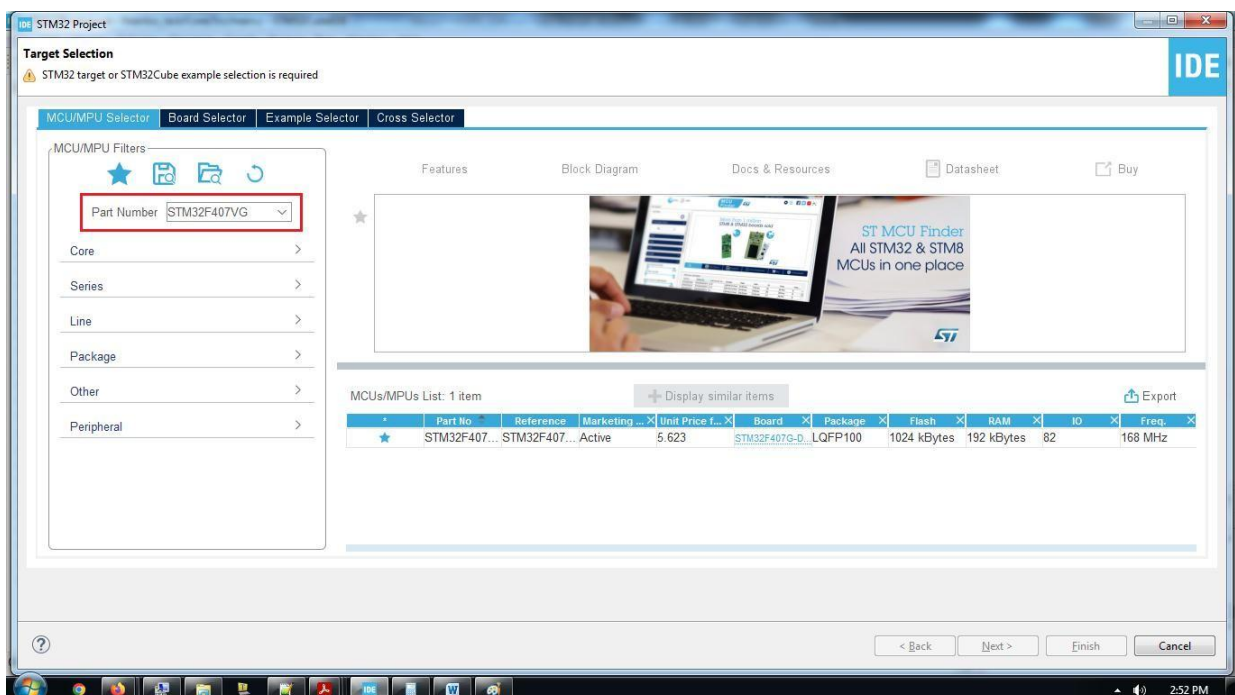
1.2. Creating a New Project.

Open the STM32CubeIDE (preferably Run as administrator)

Step 1: To create a new project, click on File→New→STM32 Project.



Step 2: Select the MCU part number.

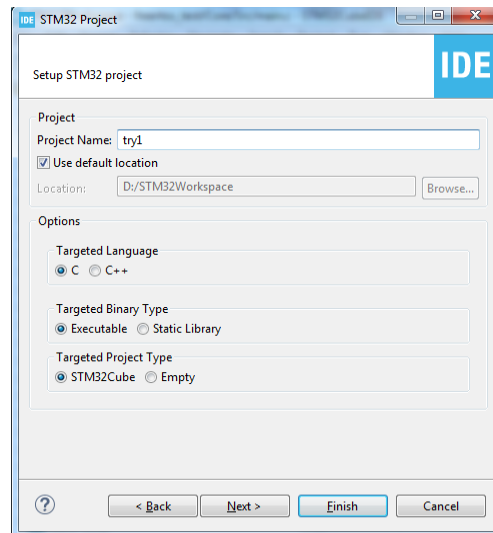


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In the Target Selection window, enter the part number **STM32F407VG** (this part is populated on the board), in the Part Number window.

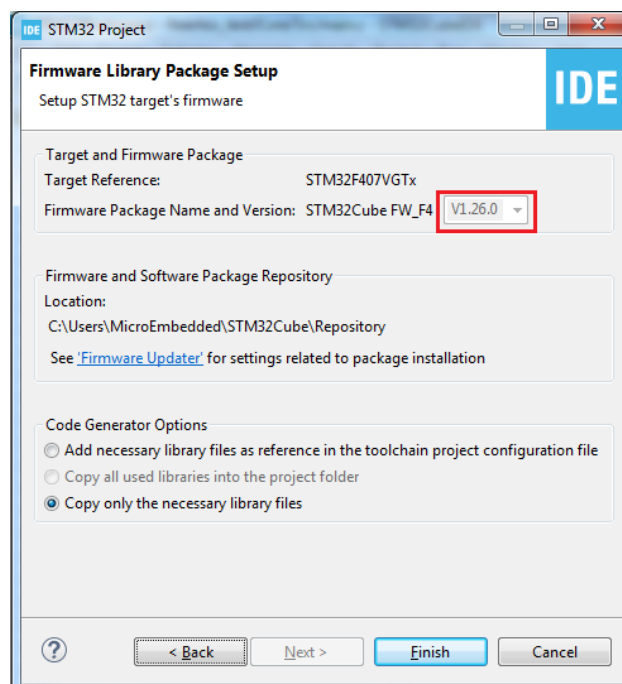
MCU/MPU Selector → Part Number.

Now select the part (click near the star ) and click Next. Step 3:



In the Project Name window, give a unique name to the project. (you cannot have two projects with the same name). Click Next.

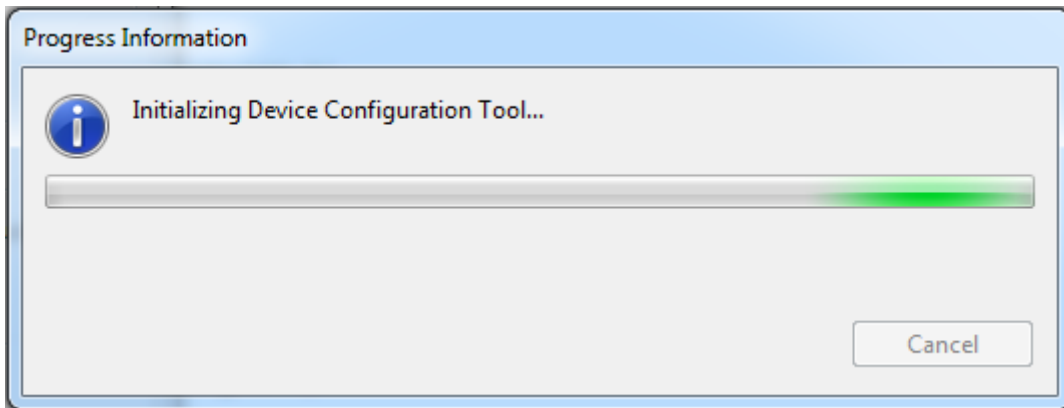
Step 4:



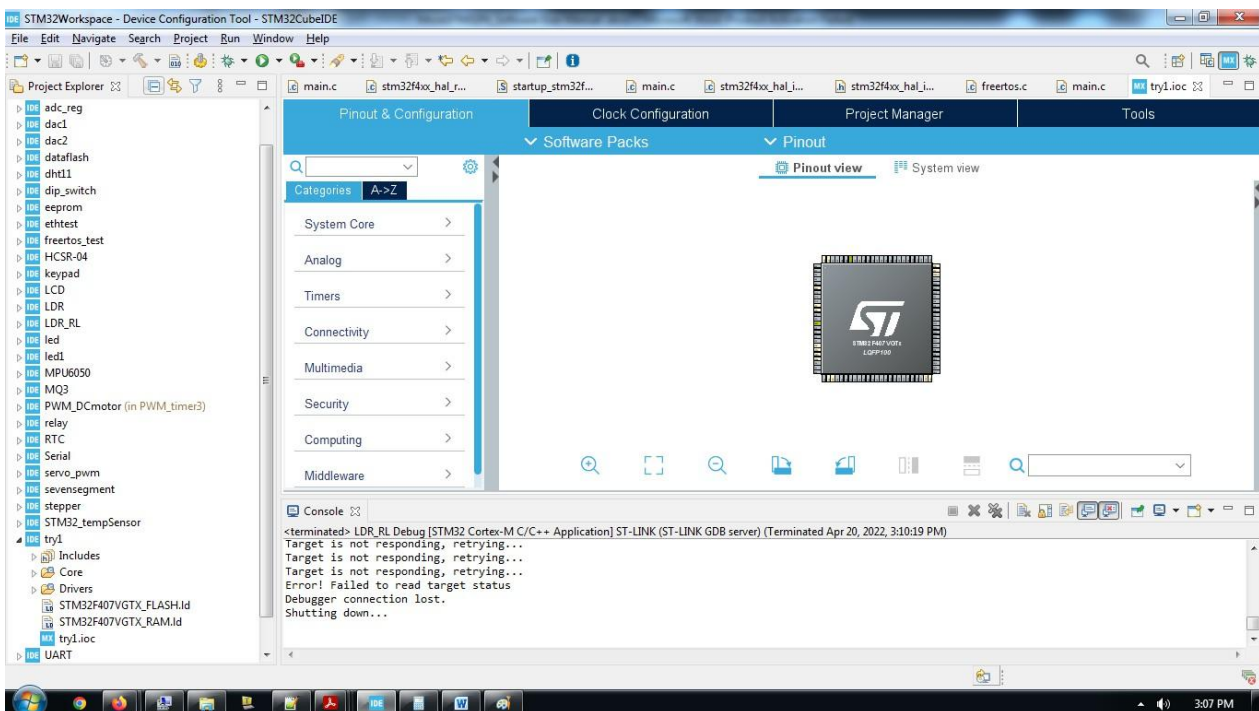
Make sure that the Firmware Package Name and Version is **V1.26.0**.

Click Finish to complete the process of creating the new project.

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After the process is complete the project Template will get created.



The Project Structure is as follows:

All the source code (.c) and header (.h) files for the HAL (library function for peripherals) are located in

C files :Project_name → Drivers → STM32F4xx_HAL_Driver → Src

.h files :Project_name → Drivers → STM32F4xx_HAL_Driver → Inc

All the source code (.c) and header (.h) files for the user program are located in C files

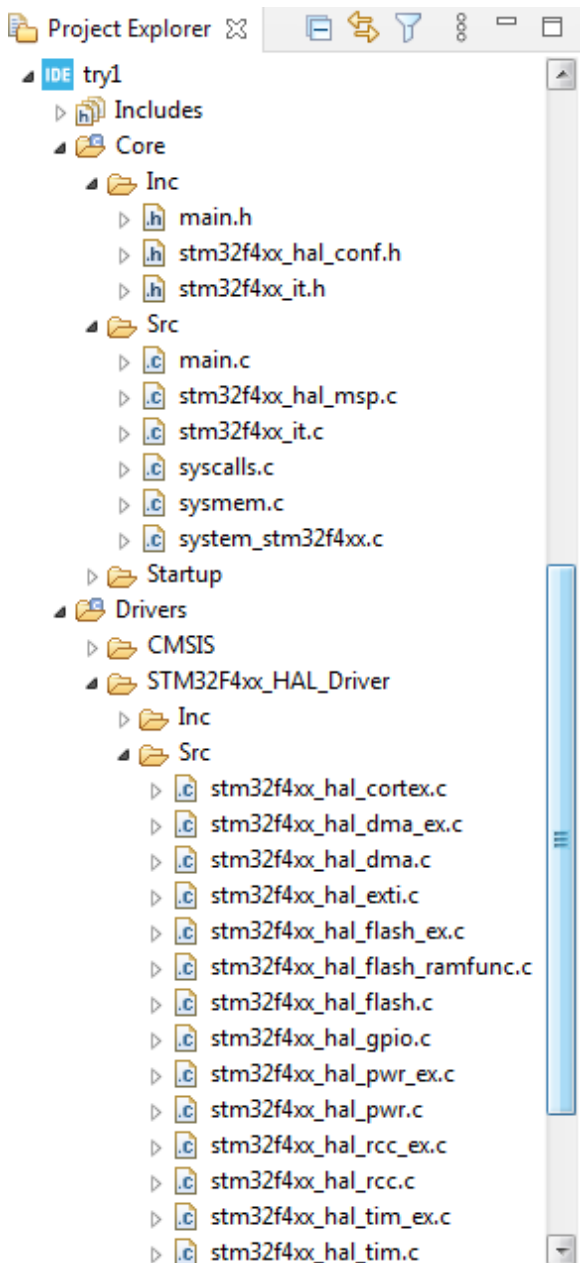
:Project_name → Core → Src

.h files :Project_name → Core → Inc

The user can write the program in main.c file.

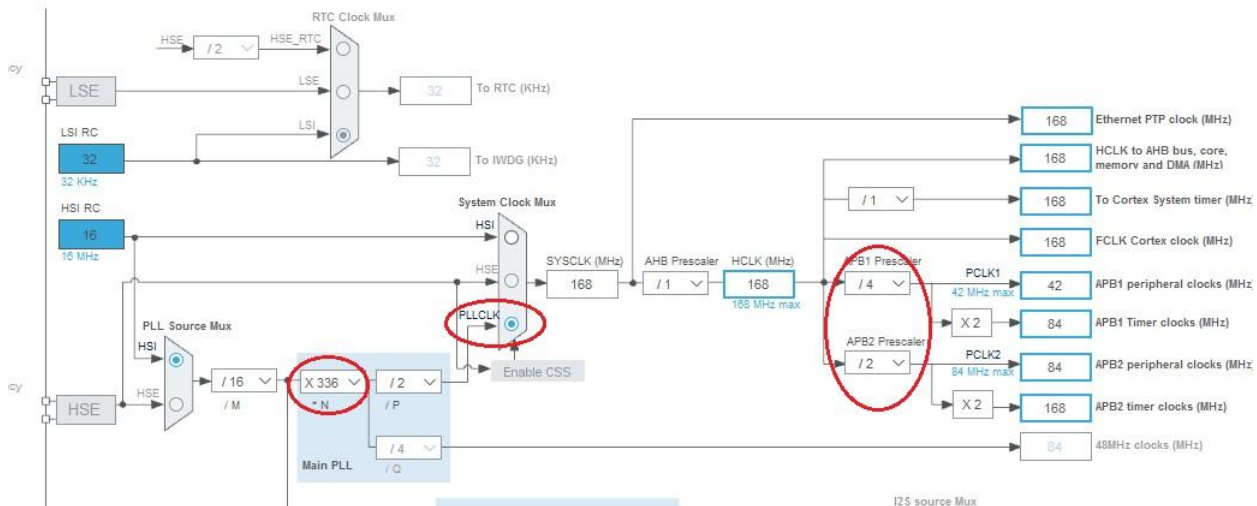
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The code for clock initialization and HAL initialization are already done and available in main.c file.



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If the user wants the maximum clock frequency of 168 MHz, the PLL should be set as follows;



In the PLL section make N = 336

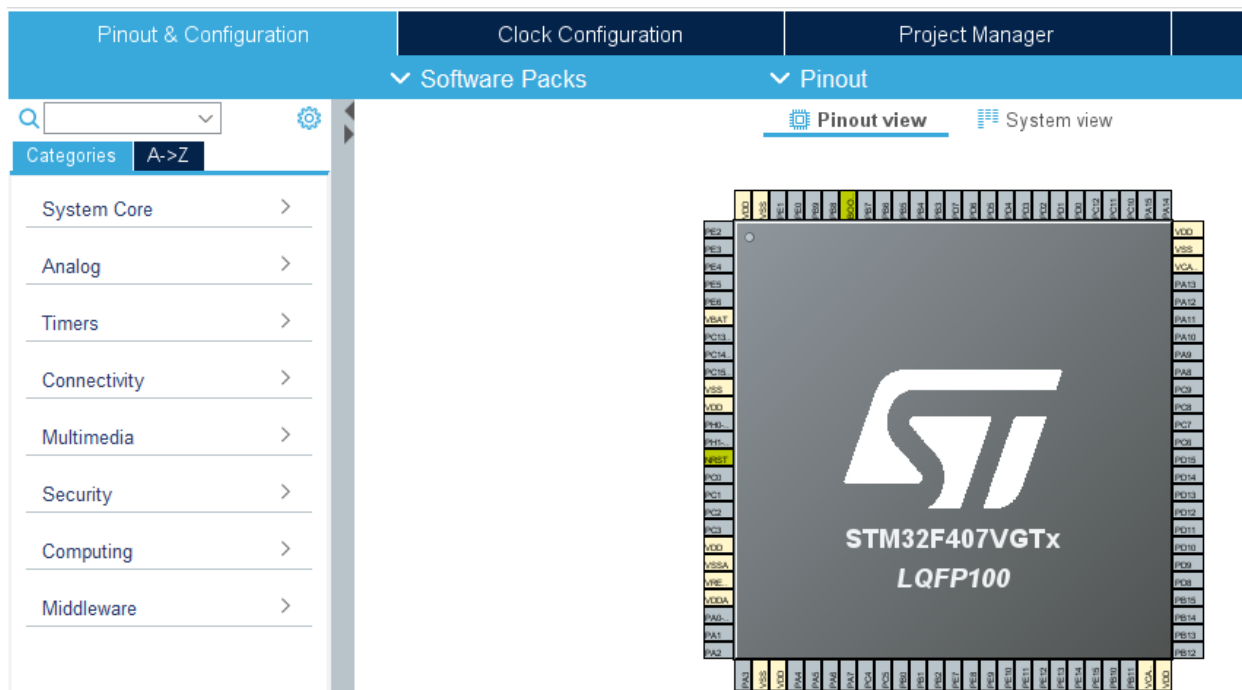
In System Clock Mux select PLLCLK In

APB1 Prescaler use /4

In APB2 Prescaler use /2

This will set all the clocks to the core and peripherals for maximum allowed frequency.

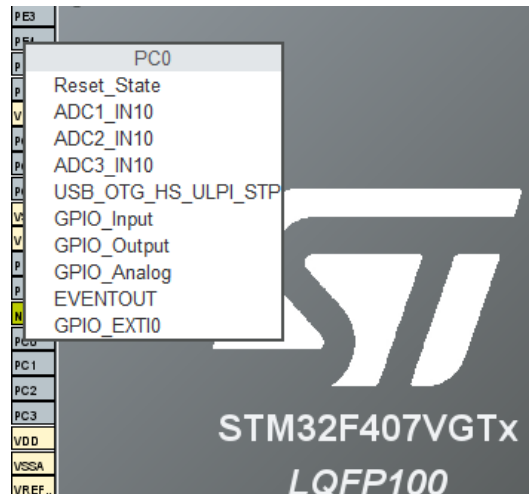
1.4. Pinout & Configuration.



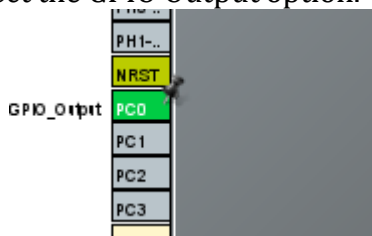
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We can use the GUI tool in the IDE to set the pin configuration.

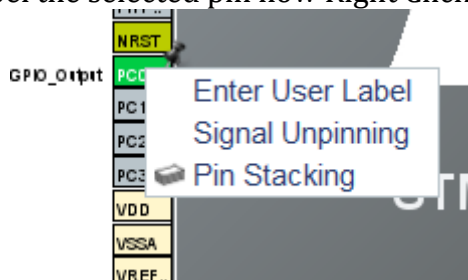
Select the pin (let's take pin PC0 which is connected to LED1). Locate the pin on the MCU pin map. Left Click on the pin PC0.



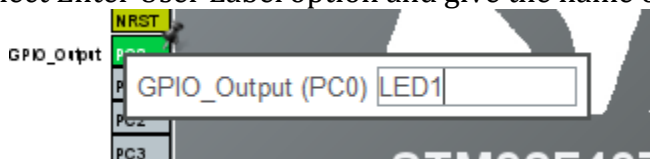
Select the GPIO Output option.



To label the selected pin now Right Click on the pin



Select Enter User Label option and give the name of your choice in the window

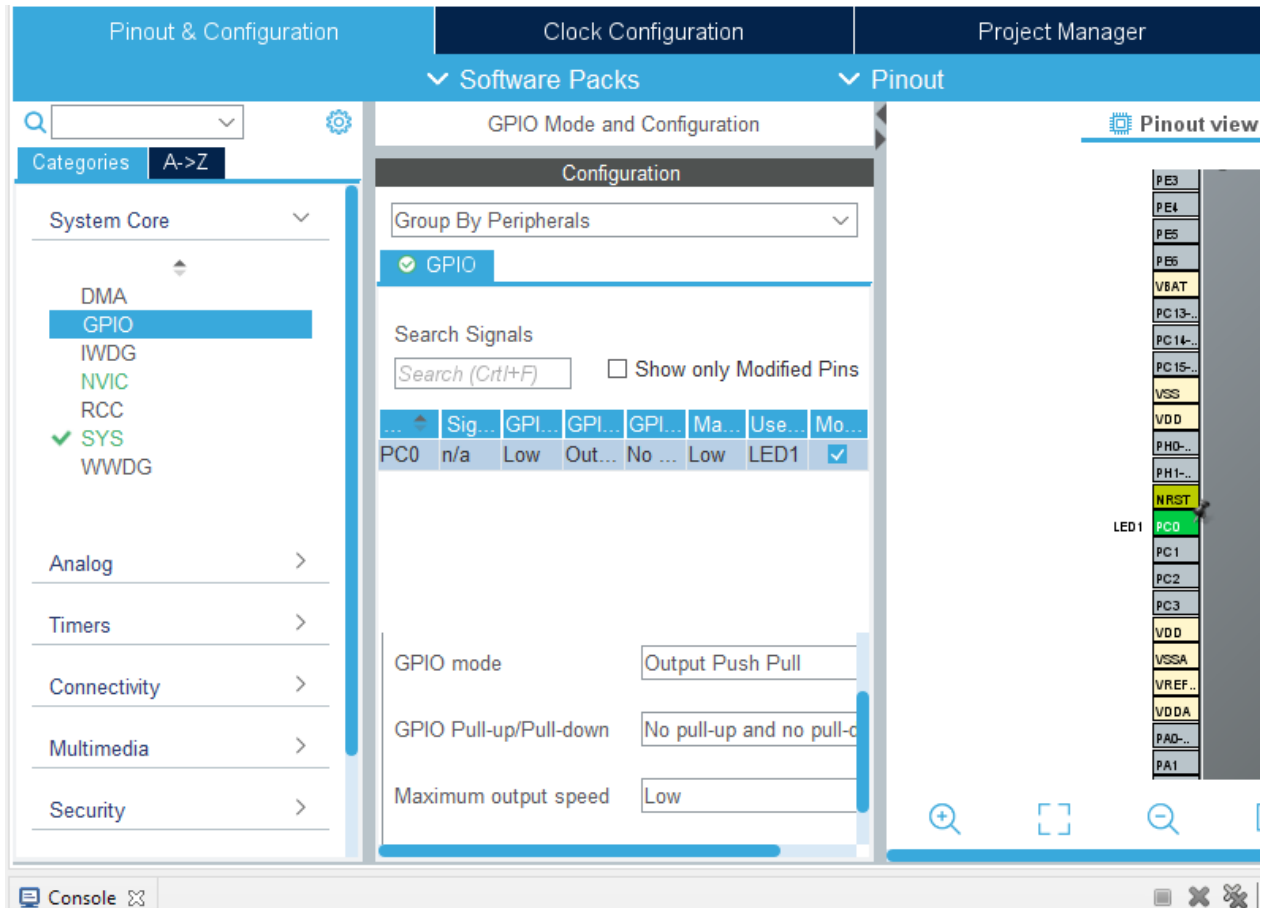


Now when you press enter, the pin name will be set to LED1. This name will be used by the library and automatic code generation tool throughout the code.

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You can also set the pin parameters from the tool.

Expand the System Core and select GPIO. The selected GPIO will be shown and below you can set the parameters for the selected Pin.



The same procedure applies to multiple pins and peripherals. To set the parameters for the peripherals, set the pins, select the peripherals from the Pinout & Configuration settings block and make appropriate changes.

After you have set the parameters, save the project. You will be prompted to generate code. When you click yes, the IDE will generate the initialization code and template in the main.c file.

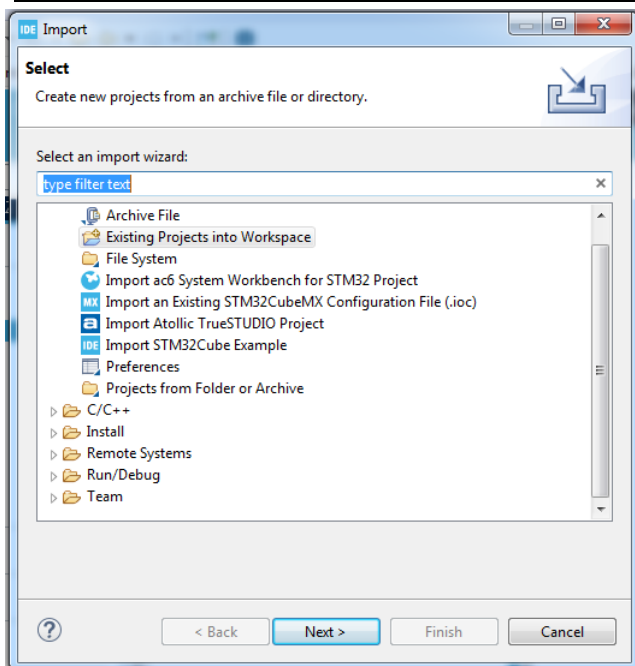


Alternatively, you can also click on the  Device Configuration Tool Code Generation to generate the code. Or go to Project → Generate Code.

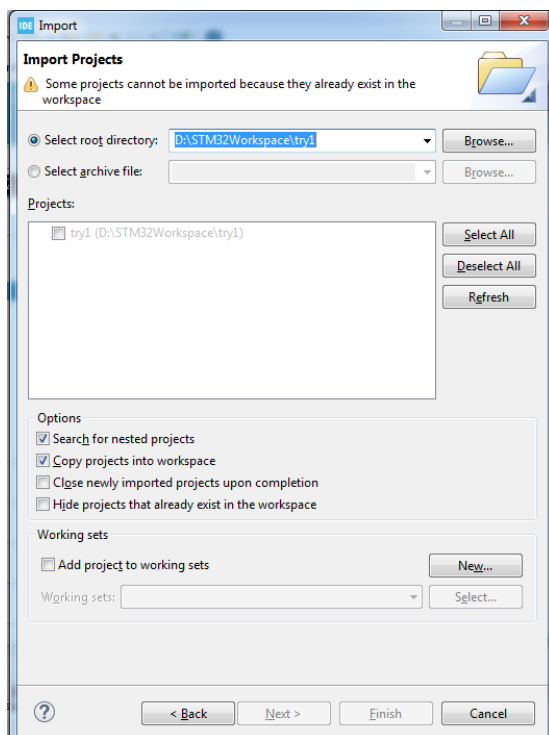
1.5. Importing an existing project in to the workspace.

When you want to import an existing project in to the workspace, click on File → Import.

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Select – Existing Projects into Workspace. Click Next.



In select root directory, browse to the folder which contains the project to be imported. Also make sure that the Copy projects into workspace field is selected in the options menu. Click Finish. The project will be imported into the workspace.

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1.6. Building / Compiling the project.

After writing the code in the main.c file compile the project by Right clicking on the project name and click Build Project.

Also you can select the project name. click on Project → Build Project.

Else you can select the project and click on  icon to build the project.

After successful build the bin file and elf file will be created. This file can be flashed into the microcontroller using the ST emulator.

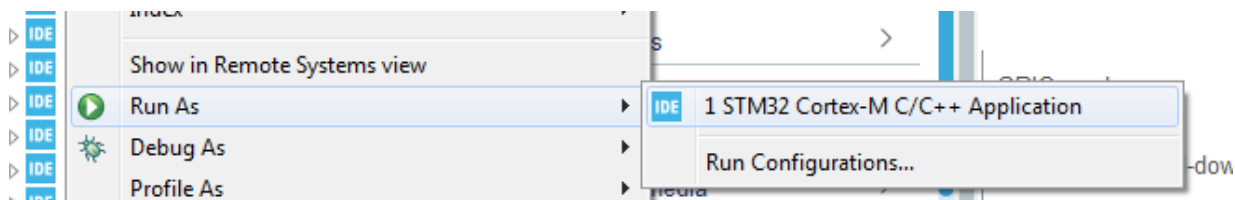
1.7. Downloading into the Microcontroller.

To download the hex file into the microcontroller, connect the Emulator to PC using the USB cable and the FRC cable to connector CN22 (JTAG Header).

Power on the board.

Make sure that Boot0 and Boot1 switches are towards GND (2-3). And switch SW13 is towards 2-3.

Now select the project, right click and select **Run As → STM32 cortex m C/C++ Application**



Also, you can select the project → Run → Run. Else

click on  icon to download the file.

Press RESET switch SW38 and observe the output.

1.8. Debug the code.

To debug the code; select the project → Right click → Debug As → STM32 Cortex-M C/C++ Application.



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Also select the project →Run →Debug.

Or select the project →click on debug icon.



The hex file will get loaded and the program will halt at main function. The IDE will enter the debug perspective and you can use the step into to step out to debug the code line by line.

CONCLUSION:
